CELL BIOLOGY SEMINAR · FIRST-YEAR MEDICINE

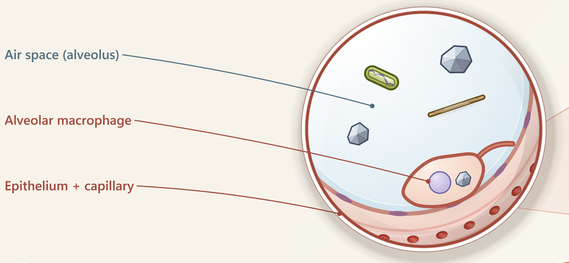
How a macrophage *destroys* what it engulfs

Phagocytosis, lysosomal biogenesis and vesicular trafficking, and what happens when particles cannot be cleared

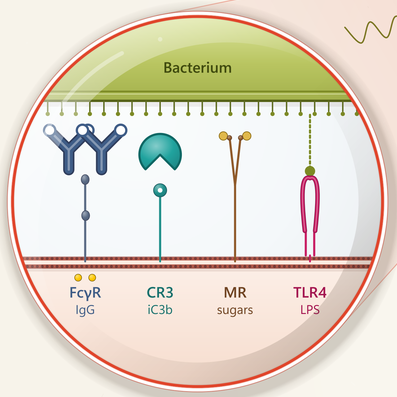
Aitor · Nerea · Urtzi · Eneritz · Nora
First-year Medicine · Cell Biology
[Faculty · University · Academic year: to complete]

1 Alveolar macrophages AITOR

- **Tissue-resident phagocytes** of the innate immune system
- Live **in the alveolar lumen**, on the epithelium, bathed in surfactant-rich lining fluid
- Clear **microbes, dust and dead cells**; their cytokines alert other immune cells



2 Recognition and engulfment NEREA

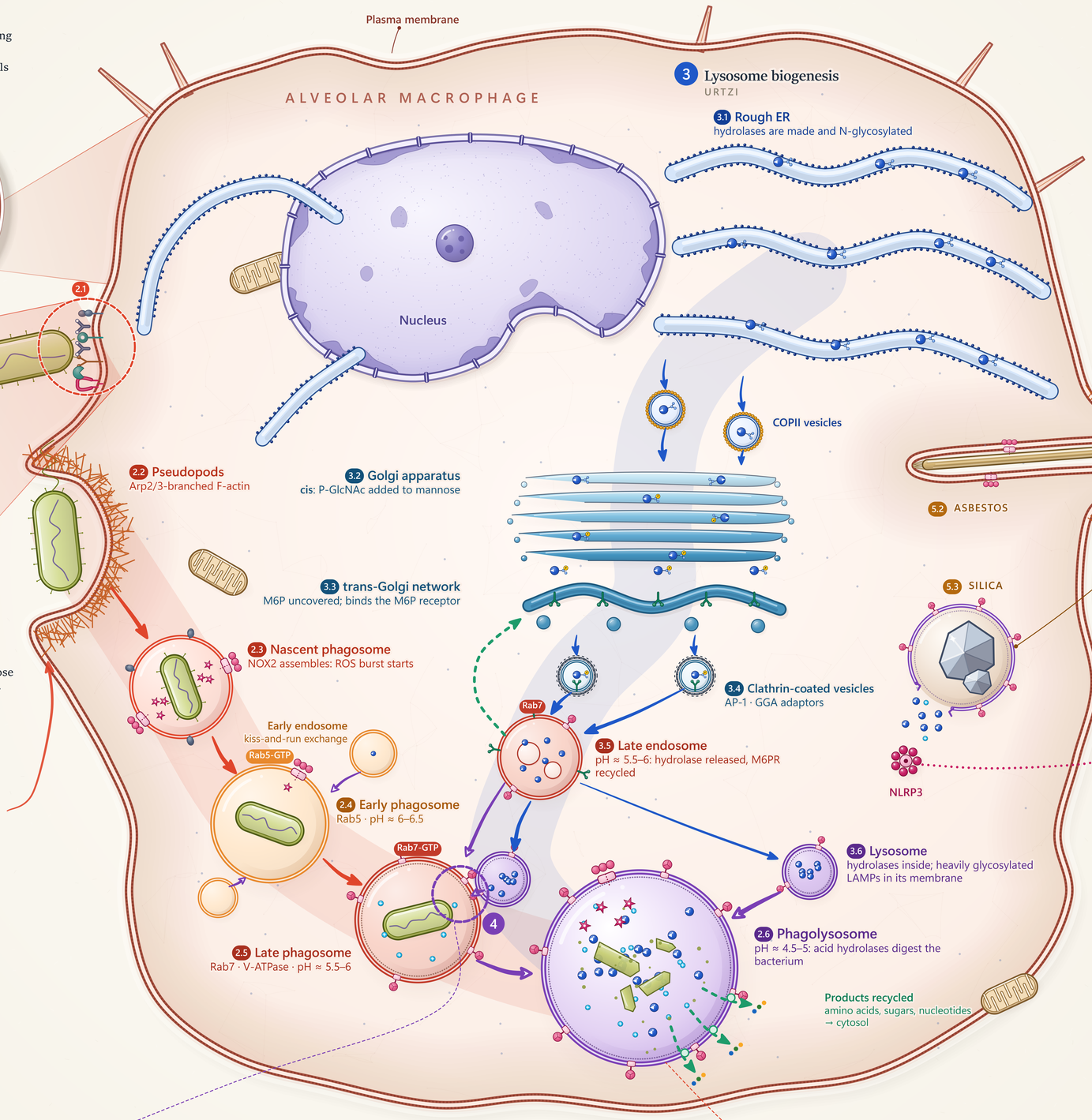


Opsonins (**IgG**, **iC3b**) bridge bacterium and receptor; FcγR, CR3 and the mannose receptor trigger uptake. **TLR4** senses LPS and only signals. CR1 binds C3b/C4b.

2.2 Engulfment: a receptor "zipper"

FcγR cluster → Syk → PI3K → Rac · Cdc42 → Arp2/3

Branched F-actin pushes **pseudopods** around the particle (2.2, the cup in the cell) until the membrane seals into a **phagosome** (2.3).



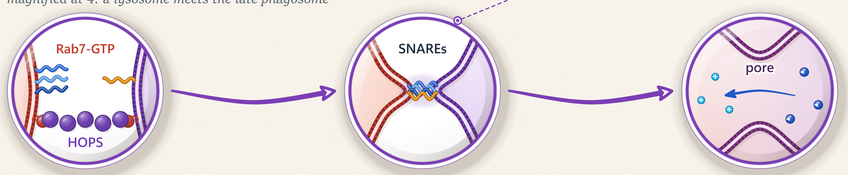
READING THE FIGURE

- Phagocytic route
- Enzyme (biosynthetic) route
- Membrane fusion
- Recycling
- Pathological effect
- Inflammatory signal
- hydrolase + M6P
- M6P receptor
- V-ATPase
- NOX2
- ROS
- H^+

Steps 2.1 → 2.6 happen in sequence, while the enzyme route 3.1 → 3.6 runs continuously alongside them.

4 Docking and fusion ENERITZ

magnified at 4: a lysosome meets the late phagosome



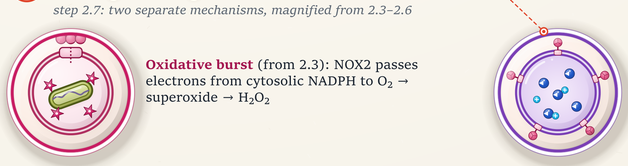
a **Tethering.** Rab7-GTP recruits the HOPS complex, which bridges both membranes

b **Zipping.** Syntaxin-7, Vti1b and syntaxin-8 coil with VAMP7

c **Fusion.** A pore opens and contents mix; NSF later recycles the SNAREs

2.7 Killing and digestion NEREA

step 2.7: two separate mechanisms, magnified from 2.3–2.6



Oxidative burst (from 2.3): NOX2 passes electrons from cytosolic NADPH to O_2 → superoxide → H_2O_2

Acid hydrolysis (after fusion, 2.6): V-ATPase pumps H^+ in; cathepsins, lipases, nucleases and glycosidases dismantle the bacterium

Products (amino acids, sugars, nucleotides) leave for reuse; some peptides are presented on **MHC II** to T cells.

5 When particles persist NORA

occupational silicosis and asbestosis



Inhaled particles reach the alveoli, where macrophages take them up but cannot digest them.

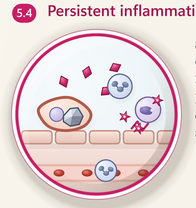
5.2 Long fibres: frustrated phagocytosis

Pseudopods spread along a fibre too long to enclose, but the cup never closes: ROS and lysosomal enzymes leak out. Short fibres can be engulfed yet still persist.

5.3 Silica: lysosomal injury

Crystals can rupture the phagolysosome membrane. Leaked cathepsins, K^+ efflux and ROS can activate the **NLRP3 inflammasome** → caspase-1 → IL-1β. Cells that die release the crystal, which is engulfed again.

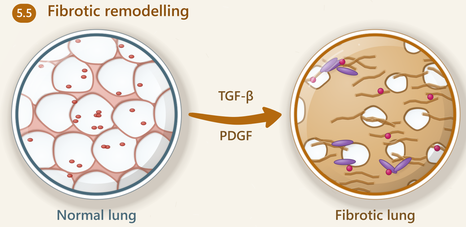
5.4 Persistent inflammation



Signalling: IL-1β, TNF-α and chemokines recruit neutrophils and monocytes.

Injury: their ROS and proteases damage the epithelium; the particle stays.

5.5 Fibrotic remodelling



Myofibroblasts deposit **collagen**: septa thicken and stiffen, air spaces shrink and gas exchange fails.

6 Take-home AITOR



Clearing a microbe needs two routes to converge: the maturing phagosome and lysosomal enzyme delivery. When the cargo cannot be degraded, the same machinery drives lung injury.

REFERENCES

1 Alberts B, Heald R, Johnson A, et al. Molecular Biology of the Cell, 7th ed. New York: W.W. Norton; 2022.
2 Hussell T, Bell TJ. Alveolar macrophages: plasticity in a tissue-specific context. Nat Rev Immunol 2014;14:81–93.
3 Flannagan RS, Jaumouillé V, Grinstein S. The cell biology of phagocytosis. Annu Rev Pathol 2012;7:61–98.
4 Kinchen JM, Ravichandran KS. Phagosome maturation: going through the acid test. Nat Rev Mol Cell Biol 2008;9:781–795.
5 Nunes P, Demaurex N, Dinauer MC. Regulation of the NADPH oxidase and associated ion fluxes during phagocytosis. Traffic 2013;14:1118–1131.
6 Braulke T, Bonifacino JS. Sorting of lysosomal proteins. Biochim Biophys Acta 2009;1793:605–614.
7 Luzio JP, Pryor PR, Bright NA. Lysosomes: fusion and function. Nat Rev Mol Cell Biol 2007;8:622–632.
8 Jahn R, Scheller RH. SNAREs: engines for membrane fusion. Nat Rev Mol Cell Biol 2006;7:631–643.
9 Hornung V, Bauernfeind F, Halle A, et al. Silica crystals and aluminum salts activate the NALP3 inflammasome through phagosomal destabilization. Nat Immunol 2008;9:847–856.
10 Leung CC, Yu ITS, Chen W. Silicosis. Lancet 2012;379:2008–2018.
11 Donaldson K, Murphy FA, Duffin R, Poland CA. Asbestos, carbon nanotubes and the pleural mesothelium: a review of the hypothesis regarding the role of long fibre retention in the parietal pleura, inflammation and mesothelioma. Part Fibre Toxicol 2010;7:5.
12 Wynn TA, Ramalingam TR. Mechanisms of fibrosis: therapeutic translation for fibrotic disease. Nat Med 2012;18:1028–1040.

Original illustration · schematic, not to scale